Practical 5: Installing KoboToolbox & Designing a Survey

Day 3 — Duration: 3 Hours

SDI Training for ICT Professionals — Climate Risk Database

Based on official kobo-install: `https://github.com/kobotoolbox/kobo-install`

What you will accomplish: By the end of this practical, you will have KoboToolbox running on your laptop, a flood assessment survey form designed and deployed, and test data submitted with GPS coordinates. KoboToolbox is a free tool for mobile data collection widely used in humanitarian and development work.

Before you start:

- **Docker** must be installed and running (Docker Compose V2 required — V1 is NOT supported)
- **Python 3.10+** must be installed
- **Git** must be installed and accessible from the terminal
- At least **4 GB free RAM** and **10 GB free disk space**
- **Port 80** must be free. If CRD is running, stop it first:
  `cd  /my_crd && docker compose down`

Part 1: Installing KoboToolbox (2 Hours)

Verify Prerequisites

Open your terminal:

- **Windows:** Open **PowerShell**
- **Mac:** Open **Terminal**
- **Ubuntu:** Press `Ctrl+Alt+T`

Step 1: Verify that Git, Python, and Docker are installed:

```
git --version
python3 --version
docker --version
docker compose version
```

You should see: Version numbers for each command. If any command says "not found":

```
# Install git (Ubuntu):
sudo apt install -y git

# Install Python 3.10+ (Ubuntu):
sudo apt install -y python3.10

# Docker: see Practical 1 or Practical 2
```

Important: `git` must be accessible from the terminal. The kobo-install script calls git internally to clone the kobo-docker repository. If git is not found, you will see this error:

```
FileNotFoundError: [Errno 2] No such file or
  directory: ''
```

This means git is not installed or not in your PATH.

Download and Run the KoboToolbox Installer

Step 2: Navigate to your home folder and clone the installer:

```
cd ~
git clone https://github.com/kobotoolbox/kobo-install.git
cd kobo-install
```

What this does: Downloads the KoboToolbox installation scripts from GitHub. These scripts automate the complex process of setting up KoboToolbox with Docker.

You should see:

```
Cloning into 'kobo-install'...
remote: Enumerating objects: ...
Resolving deltas: 100% (...), done.
```

Step 3: Start the installation wizard:

```
python3 run.py
```

What this does: The first time you run this command, it launches an interactive setup wizard. The wizard asks you questions about how to configure KoboToolbox. You type your answers and press Enter.

Answer the Setup Wizard Questions

Step 4: The wizard will ask several questions. Here are the recommended answers for a local training setup:

Wizard answers — follow these carefully:

Q1: "What kind of installation do you want?"

- Type: `1` (for "Workstation" / local machine) → Press Enter

Q2: "Where do you want to install?" (installation directory)

- Accept the default (usually `../kobo-docker`) → Press Enter

Q3: "Do you want to use HTTPS?"

- Type: `2` (for No) → Press Enter

Q4: "Super user's username?"

- Type: `admin` → Press Enter

Q5: "Super user's password?"

- Type a password (e.g. `Admin123`) → Press Enter

- (For training only — use a strong password on real servers)

Q6: "Do you want to activate backups?"

- Type: 2 (for No) → Press Enter
- (Not needed for training)

Note: The exact questions may vary slightly depending on the version of kobo-install. The instructor will guide you through any differences. The key points are: **Workstation** installation, **no HTTPS**, and create a **superuser account**.

Step 5: After answering all questions, the wizard will:

a) Clone the `kobo-docker` repository (this is where the actual Docker configuration lives)

b) Generate environment configuration files

c) Pull Docker images from Docker Hub

d) Start all containers

```
Cloning kobo-docker into ../kobo-docker ...
Pulling images ...
Creating containers ...
```

This takes 10–20 minutes depending on your internet speed.

Troubleshooting Common Errors

Step 6: If you see `FileNotFoundError: No such file or directory: '':`

This error means git is not installed or not in the system PATH. The kobo-install script needs git to clone kobo-docker.

```
# Fix: Install git
sudo apt install -y git

# Verify git works
which git
git --version

# Delete any partial config and retry
rm -f ~/kobo-install/.run.conf
cd ~/kobo-install
python3 run.py
```

Step 7: If you see "Port 80 is already in use":

```
# Check what is using port 80
sudo lsof -i :80

# If CRD/GeoNode is running, stop it
cd ~/my_crd && docker compose down

# Retry
cd ~/kobo-install && python3 run.py
```

Step 8: If you see "Docker Compose V1 is not supported":

```
# Check your Docker Compose version
docker compose version
# Must show V2.x, NOT docker-compose V1

# If using old docker-compose, upgrade Docker:
sudo apt remove docker-compose
sudo apt install docker-compose-plugin
```

Step 9: If you want to restart the setup from scratch:

```
cd ~/kobo-install
python3 run.py --setup
```

This re-runs the setup wizard without needing to re-clone.

Verify the Installation

Step 10: Check that all containers are running:

```
cd ~/kobo-install
python3 run.py --info
```

What this does: Shows information about your KoboToolbox installation including URLs and container status.

Step 11: You can also check logs:

```
python3 run.py --logs
```

Step 12: Open your browser and go to:

```
http://localhost
```

You should see: The KoboToolbox login page with the KoboToolbox logo.

If you see a blank page or error: Wait 3–5 more minutes and refresh. KoboToolbox takes time to fully initialise. If it still does not work:

```
python3 run.py --logs
```

Look for error messages in the output.

Step 13: Log in with the superuser account you created during the wizard:

- **Username:** `admin`
- **Password:** the password you chose (e.g. `Admin123`)

You should see: The KoboToolbox dashboard with a "New" button and an empty project list.

Useful kobo-install commands:

- `python3 run.py` — Start KoboToolbox
- `python3 run.py -stop` — Stop all containers
- `python3 run.py -logs` — View container logs

- `python3 run.py -info` — Show installation info
- `python3 run.py -setup` — Re-run setup wizard
- `python3 run.py -update` — Update to latest version
- `python3 run.py -help` — Show all available commands

Part 2: Designing a Flood Assessment Survey (1 Hour)

You will now design a survey form for collecting flood damage data in the field. This form uses the **visual form builder** — no coding or XLSForm knowledge needed.

Create a New Form

Step 14: On the KoboToolbox dashboard, click the blue **"New"** button.

Step 15: Select **"Build from scratch"** (or **"Build form"**).

You should see: The form builder opens with a blank area in the centre.

Step 16: At the top, click the form name field and type:

```
Flood Damage Assessment - Tanga 2026
```

Add Survey Questions

Add 7 questions to the form. For each question, click **"+"** and configure it.

Step 17: Question 1 — GPS Location:

- Select question type: **"Point"** (under Location/GPS)
- Label: `GPS Location of Assessment`

Step 18: Question 2 — Flood Depth:

- Select question type: **"Number"** (or "Integer")
- Label: `Flood Water Depth (centimetres)`
- Set minimum value: `0`, maximum: `500`

Step 19: Question 3 — Date of Flooding:

- Select question type: **"Date"**
- Label: `Date of Flooding`

Step 20: Question 4 — Ward Name:

- Select question type: **"Select One"**
- Label: `Ward Name`
- Add choices: `Chumbageni`, `Makorora`, `Ngamiani`, `Mzingani`, `Tanganyika`, `Other`

Step 21: Question 5 — Photo of Damage:

- Select question type: **"Photo"** (or "Image")
- Label: `Photo of Flood Damage`

Step 22: Question 6 — Households Affected:

- Select question type: **"Number"**
- Label: `Number of Households Affected`
- Set minimum value: `0`

Step 23: Question 7 — Type of Damage:

- Select question type: **"Select Multiple"** (checkboxes)
- Label: `Type of Damage Observed`
- Add choices: `Structural damage to buildings`, `Road damage or blockage`, `Crop or farmland damage`, `Contaminated water supply`, `Displacement of residents`, `Loss of personal property`

Step 24: Review your form. All 7 questions should be listed in order.

Deploy the Form

Step 25: Click **"Save"** (top-right) to save your form.

Step 26: Click **"Deploy"** (may show as a rocket icon or "Deploy this version").

Step 27: Confirm deployment. The form status should change to **"Deployed"**.

Submit Test Entries

Step 28: Click **"Open"** or the Enketo web form link to open the survey in your browser.

Step 29: Fill in **Test Entry 1**:

- **GPS:** Click the map and place a point near Tanga: Lat **-5.069**, Lon **39.096**
- **Flood Depth:** `45`
- **Date:** Today's date
- **Ward:** `Chumbageni`
- **Photo:** Skip or take any photo
- **Households:** `12`
- **Damage:** Check `Structural damage` and `Road damage`

Click **"Submit"**.

Step 30: Submit **Test Entry 2** with different values (different ward, different GPS point slightly north at **-5.060**, **39.100**).

Step 31: Submit **Test Entry 3** with your own values.

View the Collected Data

Step 32: Go back to the dashboard. Click your form name.

Step 33: Click the **"Data"** tab. You should see a table with all 3 submissions.

Step 34: Explore the data views:

- **Table view:** Spreadsheet-like data table
- **Map view:** GPS points displayed on a map near Tanga

- **Reports:** Summary charts and graphs

Checkpoint

Before proceeding to Practical 6, confirm:

☐ **KoboToolbox running:** Login page loads at `http://localhost`

☐ **Admin login works:** You can sign in with your admin account

☐ **Form created:** "Flood Damage Assessment - Tanga 2026" exists

☐ **Form deployed:** Status shows "Deployed"

☐ **Test data submitted:** 3 entries visible in the Data tab

☐ **Map view works:** 3 points appear near Tanga on the map

Excellent work! You now have a complete field data collection system. In the next practical, you will export this data and upload it to CRD — connecting field collection to your spatial data platform.

End of Practical 5 — Proceed to Practical 6: From KoboToolbox to CRD